Geography of Energy, Vitality, and Happiness Across US Counties (from X/Twitter data)

Tuesday 3rd November, 2026 14:04

We use X (Twitter) data to study energy, vitality, and happiness Across US Counties. Energy, vitality, and happiness is measured based on human expression on X. We use 3 metrics measuring energy/vitality: having energy, positive functioning, and vitality; and 6 measuring happiness: life satisfaction, happiness, peace with thoughts and feelings, expressing optimism, purpose in life, and having hope. Cronbach's Alpha scores of 0.85-0.92 indicate high internal consistency and reliability of our novel scales. This is the first study of geography of energy/vitality, where we find a very clear urban-rural gradient with energy/vitality raising from low in large counties to high in small counties. Energy/vitality may be actually more important than happiness or at least as important, and it is striking that we know nothing about its geography. There is literature, on the other hand, on geography of happiness–cities tend to be less satisfied with life. We confirm this finding, but overall on an index and other happiness measures, we find an important qualification. It is not the very smallest counties in the 1st population decile that are the happiest, but small counties in deciles 2-4 (5.1-18.5k). The very smallest counties in 1st decile (<5.1k) are actually the least happy along with the largest counties in 10th decile (>205k). We also zoom in on 20 largest US counties, and 100 top/bottom counties on our wellbeing metrics. While X wellbeing expressions tend correlate moderately at about .6 with survey data, operationalizations have limitations and geotagged tweets over-represent sub-populations.

1 Introduction

Terms SWB, happiness, and life satisfaction are usually used interchangeably in happiness and social indicators literatures. But specifically they mean rather cognitive global subjective self-evaluation of one's life as a whole rather than momentary affective happiness. Here it rather is the latter, momentary affective happiness, as human expression is recorded via tweets on X.[1]

Much ink has been spilled on geography of Subjective Wellbeing (SWB), including in the US, in particular there are studies across counties (e.g., Okulicz-Kozaryn and Mazelis 2018, Winters et al. 2015, Okulicz-Kozaryn et al. 2024), and across urban rural continuum (Okulicz-Kozaryn 2024, 2023, Okulicz-Kozaryn and Valente 2020). Major findings have been that humans are less satisfied with their lives in populated dense North East, but also in poorer Midwest, and some of the lowest life satisfaction scores in addition to largest cities such as NYC and Philadelphia, have been recorded in South, in particular in poor Alabama, Mississippi, and Louisiana. Greatest life satisfaction, on the other hand, has been found in sparsely populated North and West, in states such at Nebraska and Utah, but not so much in overpopulated California.

These were happiness and geographic literatures. One limitation of these literatures is sole use of survey data such as GSS and BRFSS and typically just one survey item about life satisfaction such as "How satisfied are you with your life as a whole?" But human wellbeing can be measured in many other ways. Here we use across about 3k US counties 9 metrics: 6 measuring happiness, and 3 measuring energy/vitality. This is the main contribution of our study to expand on measurement of county and urban-rural wellbeing. In the process of doing so we also contribute by connecting happiness and geography literatures (summarized in Okulicz-Kozaryn 2023, Okulicz-Kozaryn and Valente 2020)[2] with X literature, which seems to be mostly in computer science.

[1] Reliability and validity of twitter data discussion is postponed to section "Data."

[2] For elaboration see Ballas and Rijnks (2023), Ziogas et al. (2023), Weckroth et al. (2022), Ziogas et al. (2020), Ballas (2013), Knies (2026), Sørensen (2024, 2014), Lenzi and Perucca (2022, 2020, 2016c,a,b).

2 X geography of wellbeing literature

There is a quite separate X (Twitter) literature, that started around 2010 shortly after Twitter launched in 2006. One early example on overall geography of X is widely cited Leetaru et al. (2013), but it does not study happiness nor energy/vitality nor any other wellbeing metrics. Various subsequent studies did measure broadly defined wellbeing including energy/vitality and happiness, but they differ in important ways from our study except pioneering Mitchell et al. (2013) that we discuss towards the end of this section.

There is an interesting and related doctoral dissertation (Gupta 2018), but analyzes wellbeing geographies in UK. Chai et al. (2023) produced "twitter Sentiment Geographical Index (tSGI), a location-specific expressed sentiment database with SWB implications", but it doesn't really analyze energy/vitality nor happiness and has a global, not the US, focus.

There are neighborhood level studies in NYC studying happiness and poverty (Tan and Guan 2021) and proximity to parks (Plunz et al. 2019). And one in Detroit analyzing happiness and neighborhood conditions (Park et al. 2021). And another analyzing happy tweets, healthy food references, and physical activity across neighborhoods in Salt Lake, San Francisco, and New York (Nguyen et al. 2016). Another study analyzes sentiment across spatio-temporal dimensions (land use and times of the day) across Massachusetts (Cao et al. 2018).

Then there are several studies that are closer to ours. Hollander and Renski (2022) studies cities, like us, but unlike us excludes rural areas, and only analyzes 50 declining cities matched with 50 growing/stable cities using only 300k tweets.

Even though Hu et al. (2019) may seem similar to our study as it broadly asses sentiment and positive emotions across us counties, it is quite different. Their approach and framing are very different. Rather than focusing on geography of wellbeing like us, they explore tweeter topics and correlations in space. Rather than energy/vitality or happiness, they study various topics such as birthdays, parties, and great time on Sunday and focus on spatial correlations rather than levels.

The closest to what we do is Mitchell et al. (2013)–its focus is on geography within the US, but they study states (we study counties), and while they also study some cities like us, it is a limited set. Our data covers virtually all of the US including rural counties. They only have 80m words generated in 2011 on X, and we have >1b tweets 2010-2019. While they also show some results for all of the US in their map in figure 5, it is difficult to read for less populated areas. Another limitation that we improve on is the use of multiple energy/vitality and happiness metrics as opposed to just one happiness metric in their study. Also note the Atlantic article that discuss the arxiv version of their study.[3] We discuss Mitchell et al. (2013) more as we present our results. Note that Mitchell et al. (2013) is one of the pioneering studies done over 10 years ago.

Last, but not least there is an arxiv pre-print (Iacus and Porro 2025) by producers of the data that we use. We both study US geography of happiness, and we both use life satisfaction and happiness metrics. But here similarities end. In addition to life satisfaction and happiness metrics we use 7 other measures. Iacus and Porro (2025) measure rural-urban with USDA ERS 9 Rural-Urban Continuum Codes; we use population deciles. Iacus and Porro (2025) doesn't focus on any US counties, except 23 most rural counties in their table 4; we focus on largest 20 counties instead. Finally, Iacus and Porro (2025) in addition to geography also focus on politics and income, while we focus just on geography and dig deeper. Thus, despite using 2 variables from the same dataset, our studies differ in critical ways.

Curiously Kjell et al. (2023) shows a map similar to our county estimates in fig 5.1C and cites Jaidka et al. (2020). Jaidka et al. (2020) focuses on comparison of twitter happiness with Gallup data but doesn't analyze by urban-rural–and county-level estimates cited in Kjell et al. (2023) are not there, not even in supplemetary material, hence it is difficult to discuss. But we discuss more when discussing our results. A useful contribution of Jaidka et al. (2020) is a finding that wellbeing X measures correlate well at about .6 with survey data. Another study by Jaidka (Jaidka et al. 2021), like us, analyzes of US counties and urban-rural, but in terms of stress.

Overall, while there is X geography of happiness literature, it either focuses on different geographies (other countries, or other aggregations such as state or neighborhoods) and different (e.g., stress) or more limited metrics such as one measure of happiness

[3] https://www.theatlantic.com/technology/archive/2013/02/the-geography-of-happiness-according-to-10-million-tweets/273286/

only.

3 Data

Data are from `https://dataverse.harvard.edu/file.xhtml?fileId=13094135&version=3.1` and we use file `flourishingCountyYear` Description of the data is at `https://arxiv.org/pdf/2511.03915` (Iacus et al. 2025):

> Here we introduce the Human Flourishing Geographic Index (HFGI), derived from analyzing approximately 2.6 billion geolocated U.S. tweets (2013-2023) using fine-tuned large language models to classify expressions across 48 indicators aligned with Harvard's Global Flourishing Study[...](Iacus et al. 2025, p. 1)

HFGI indictors are about expressed wellbeing, not a survey-like measurement of well-being. These indicators capture expressions on Twitter/X.

We cut out 2020-2023 due to Covid19 pandemic, but that still leaves us with over 1b tweets. Pandemic affected US geographies very differently and thus would be likely to bias our results–low wellbeing would have been attributed to place, while it was due to Covid19. We also dropped county-year-variable cells with <5 valid tweets. If there were very few tweets in a cell, the county-year-variable average could be often more extreme than average from larger and more representative sample and thus bias disproportionately the results.

We averaged annual scores equally to one score per county over 2013-2019. For sample description see Appendix.

One could frown upon Twitter data, but here there are actually reasons to be optimistic. There are 9 wellbeing measures, and data covers 2012-2019, having it collapsed for each county over 8 years makes sample size bigger and provides more reliability.

The reliability/validity seems acceptable, again wellbeing X measures correlate at about .6 with survey data (Jaidka et al. 2020). But reliability/validity it probably not high. While some are excited and overly positive (e.g., Karamouzas et al. 2022), others raise important points specifically regarding geotagged tweets.

Geotagged users[4] are not representative of the US population. Number of geotagged users is predicted by younger users, users of higher income, and users in urbanized areas, Hispanic/Latino users and Black users, and east or west coast (Malik et al. 2015). Very important criticism comes from Jensen (2017) who directly critiques Mitchell et al. (2013) that is very close to our study:

> [...]main issues identified are inferential over-extensions resulting in over-claiming, limitations in the operationalization of key concepts, de facto sampling bias, and a failure to account for the inherent shortcomings of this form of secondary data[...]

We dutifully acknowledge the criticism and find it valid and well-taken and these limitations need to be kept in mind when interpreting the results. But again, we use a $>$100x larger sample of $>$1b than Mitchell et al. (2013) 10m. In general, the reliability/validity seems acceptable. And we use multiple measures with high internal consistency as described below.

We don't get here into twitter data processing and technicalities of natural language processing–it is not our area and we didn't do this–again, we just downloaded pre-processed data and the documentation is in Iacus et al. (2025). We want to focus here on the meaning of the variables instead.

3.1 From Twitter data to HFGI indicators

The HFGI was obtained by applying a fine-tuned large language model to geolocated tweets from the Harvard CGA Geotweet Archive. The model was trained to answer questions such as "Does this tweet express vitality?" or "Does this tweet express depression?" For

[4]For geographic analysis such as ours across counties and urban-rural spectrum one needs geographic location, i.e., geotagged tweet.

each dimension, its presence and level "low, medium, or high" were identified. These classifications were then converted into numerical scores and aggregated into county-year indicators, each ranging from -1 to 1. The resulting measures capture expressions of specific dimensions of wellbeing in geographically located tweets. Further methodological details and validation experiments are reported in the HFGI data descriptor (Iacus et al. 2025).

3.2 Construction of composite indicators

It is important to note a caveat: in the choice of variables we mostly just used our judgment, there little literature or agreed upon set of metrics here. But note that the variables were constructed so they correspond to human flourishing (Iacus et al. 2025, 2015, Curini et al. 2014). Energy/vitality scale is completely new but the items are conceptually related, and also mathematically related with Cronbach's alpha. But there is literature motivating some of our individual choices regarding happiness scale `swbI`: `Life satisfaction` and `happiness` are clearly close constructs (Michalos 1980, Song and Zhao 2023). `Purpose` and `hope` are easy to motivate–there is a lot of research. But `innerpeace` and `optimism` are more difficult, still `innerpeace` may be an essential faucet of wellbeing (Carreno et al. 2021), and `optimism` correlates with various wellbeing dimensions at about .5 (Shah et al. 2026). `Purpose` is closely related to or rather even part of wellbeing (Ryff and Singer 2008, Zika and Chamberlain 1992, Kalaidis 2014, Manson 2015, DePaulo 2021, Ryff 1989, 2014). `Hope` like optimism is closely related to wellbeing or even a part of it (Manson 2019, Pleeging et al. 2021).

We are encouraged by high Cronbach's Alpha scores that indicate high internal consistency and reliability or our novel scales. We made indices by simply averaging across indicators–these variables go very well together at high Cronbach's alpha: we have 9 indicators (3 energy/vitality and 6 happiness). 3-item energy/vitality index `eneI` has a high (good) Cronbach's alpha of 0.85. 6-item happiness index `swbI` has a very high (excellent) Cronbach's alpha of 0.92.

$$'eneI' = mean('posfunct','vitality','energy') \tag{1}$$

$$'swbI' = mean('lifesat','happiness','innerpeace','optimism','purpose','hope') \tag{2}$$

- Positive functioning. Feeling capable and effective (`posfunct`).
  low: "Everything feels like too much lately."
  med: "Getting back into a productive rhythm again."
  high: "Handled a dozen tasks today–completely in the zone!"

- Vitality. Feelings of strength and activity (`vitality`).
  low: "Still recovering from a cold, running on empty."
  med: "Tired but determined to finish this week strong."
  high: "The energy today is unreal–everything feels possible!"

- Having energy. Reports of feeling (or not) energetic (`energy`).
  low: "Can't seem to get moving this morning."
  med: "A little sluggish but getting things done."
  high: "So full of energy I could run another mile!"

- Life Satisfaction. Satisfaction with one's life as a whole (`lifesat`).
  low: "My living situation is exhausting and unfair."
  med: "Things are slowly falling into place–can't complain too much."
  high: "Watching the sunset from my porch tonight, feeling truly content."

- Happiness. The text expresses some level of happiness (`happiness`).
  low: "Mondays always feel endless and gray."
  med: "Trying to notice at least one small thing that makes me smile today."
  high: "We reached our team goal today–feeling genuinely proud and joyful!"

- Expressing optimism. Optimism about one's condition in the medium run (`optimism`).
  low: "I don't expect much to improve soon."
  med: "Maybe next month will bring better news."
  high: "Things are finally turning around–I can feel it!"

- Peace with thoughts and feelings. General feeling of calm acceptance (`innerpeace`).
  low: "Can't quiet my thoughts enough to rest."
  med: "Learning to take things as they come."
  high: "Feeling calm and centered after a long day."

- Purpose in life. Acting with a sense of purpose (`purpose`).
  low: "I'm not sure what I'm doing any of this for."
  med: "Trying to find direction and meaning in small steps."
  high: "Every action today felt aligned with what I believe in."

- Having hope. Hope about the future (`hope`).
  low: "Hard to see a way forward."
  med: "Things might improve if we stay patient."
  high: "Tomorrow will be brighter–I truly believe it."

4 Results

4.1 Urban-rural gradient

It is hard to believe for us urban America, that there exists rural America (Hanson 2015), and that in fact there are about 300 counties (yes whole counties not just villages) with population $<$ 5k.

We start with urban-rural happiness gradient (Berry and Okulicz-Kozaryn 2011) in table 1 looking at energy/vitality. There is a remarkable consistency: the urban rural happiness gradient is raising from its lowest levels in largest counties (color coded with red) to highest levels in smallest counties (green). The only one exception is the `energy` column that only varies across urban-rural at small 0.02 at shows highest level in mid sized counties. An important pattern is that it is not only the very largest counties in the top 10th decile in last row that have lowest energy/vitality, but also other large counties in 7th-9th deciles (36.7k-204.9k). So lowest energy/vitality is not only in the very largest counties in 10th decile, but already in 7th decile and higher (>36k). Likewise highest energy/vitality is not only in the very smallest counties in 1st decile (<5.1k), but already in 3rd decile and below (<13.5k), and especially in 2nd decile and below (<8.9k).

One nuance is that `posfunct` and `vitality` columns show the lowest levels (dark red) not in the very largest counties in the last row "204.9k-10077.3k," but in slightly smaller but still above the median population size rows.

Also note quite large substantive differences. Energy/vitality index `eneI` has a 12% (.35/.31) difference between 1st and 10th deciles. `Posfunct` and `vitality` are even bigger percentage differences.

Table 2 repeats the exercise but for happiness index `swbI` and individual measures. Here as expected and shown in the literature (Berry and Okulicz-Kozaryn 2011) in second colored column `lifesat`, life satisfaction raises from its lowest level for largest counties

population 2015 decile	eneI	posfunct	vitality	energy
88-5.1k	0.35	0.37	0.26	0.43
5.1k-8.9k	0.35	0.35	0.26	0.43
8.9k-13.5k	0.34	0.33	0.25	0.43
13.5k-18.5k	0.33	0.32	0.24	0.43
18.5k-25.6k	0.32	0.31	0.22	0.42
25.6k-36.7k	0.32	0.31	0.21	0.43
36.7k-52.9k	0.31	0.30	0.20	0.42
52.9k-92.6k	0.31	0.30	0.20	0.42
92.6k-204.9k	0.31	0.29	0.20	0.42
204.9k-10077.3k	0.31	0.30	0.21	0.42

Figure 1: Energy/vitality index and its dimensions. Note we stick to `lifesat` tag (life satisfaction) as in the original dataset, but again it is rather momentary affective happiness than global cognitive evaluation of one's life as a whole, life satisfaction term used in SWB literature. Estimates with 95% CI are in Appendix.

in last decile of population to its highest level for the smallest counties in first decile of population. But this pattern is only observed for this one column.

Overall, in all other columns the pattern tends to be of lowest wellbeing in 1st decile, opposite to what expected from "urban-rural happiness gradient hypothesis" (Berry and Okulicz-Kozaryn 2011), but probably as expected based on the "left behind ruralities hypothesis."

Still, the urban-rural happiness gradient hypothesis (Berry and Okulicz-Kozaryn 2011) somewhat holds, as the counties with highest wellbeing scores are still relatively small counties in 2nd-6th population size deciles (5k-37k), and some of the lowest wellbeing scores are also in the second highest 9th and 10th population deciles.

Overall, we find lowest happiness in especially 1st, but also in 10th population deciles and highest in 2nd-6th population deciles. While as expected smaller 2nd-6th deciles counties tends to be happier and largest 9th-10th deciles counties tend to be less happy, it is especially the very smallest counties in 1st deciles that are the least happy.

population 2015 decile	swbI	lifesat	happiness	innerpeace	optimism	purpose	hope
88-5.1k	0.24	0.15	0.24	0.13	0.15	0.38	0.40
5.1k-8.9k	0.27	0.14	0.27	0.16	0.19	0.41	0.43
8.9k-13.5k	0.27	0.12	0.29	0.16	0.21	0.42	0.44
13.5k-18.5k	0.27	0.11	0.29	0.16	0.21	0.40	0.43
18.5k-25.6k	0.26	0.10	0.29	0.15	0.21	0.39	0.43
25.6k-36.7k	0.26	0.09	0.30	0.15	0.22	0.39	0.44
36.7k-52.9k	0.25	0.08	0.29	0.14	0.20	0.37	0.42
52.9k-92.6k	0.25	0.08	0.29	0.14	0.20	0.37	0.43
92.6k-204.9k	0.24	0.07	0.28	0.13	0.20	0.36	0.42
204.9k-10077.3k	0.24	0.06	0.28	0.13	0.19	0.35	0.40

Figure 2: Happiness index and its dimensions. Estimates with 95% CI are in Appendix.

`Lifesat` is as expected in the literature, and do note huge urban-rural happiness gradient change all the way from .06 for 10th decile to .15 for the 1st decile, >2x difference. But `happiness` is almost the other way round from lowest in smaller counties to larger in larger counties (but not very largest), slightly an inverted U-shape with maximum at 6th decile (25.6k-36.7k) and `optimism` is very similar. `Hope` has a clear inverted U-shape with maximum at 6th decile and minimums at 1st and 10th deciles. Still, overall from the happiness index `swbI` (and `innerpeace`) it is not the very smallest ones like <5.1k, but also small ones already at <18.5k.

There are also some nuances, for instance 6th decile (25.6k-36.7k) scores highest on `happiness`, `optimism`, and `hope`.[5] Hence,

[5]Along with 3rd decile.

for these measures there is evidence of inverted U-shape: highest happiness is around median population size and lowest in smallest and largest counties.

4.2 Across counties

We turn to mapping[6] energy/vitality index `eneI` and happiness index `swbI` across counties. `EneI` is in figure 3.

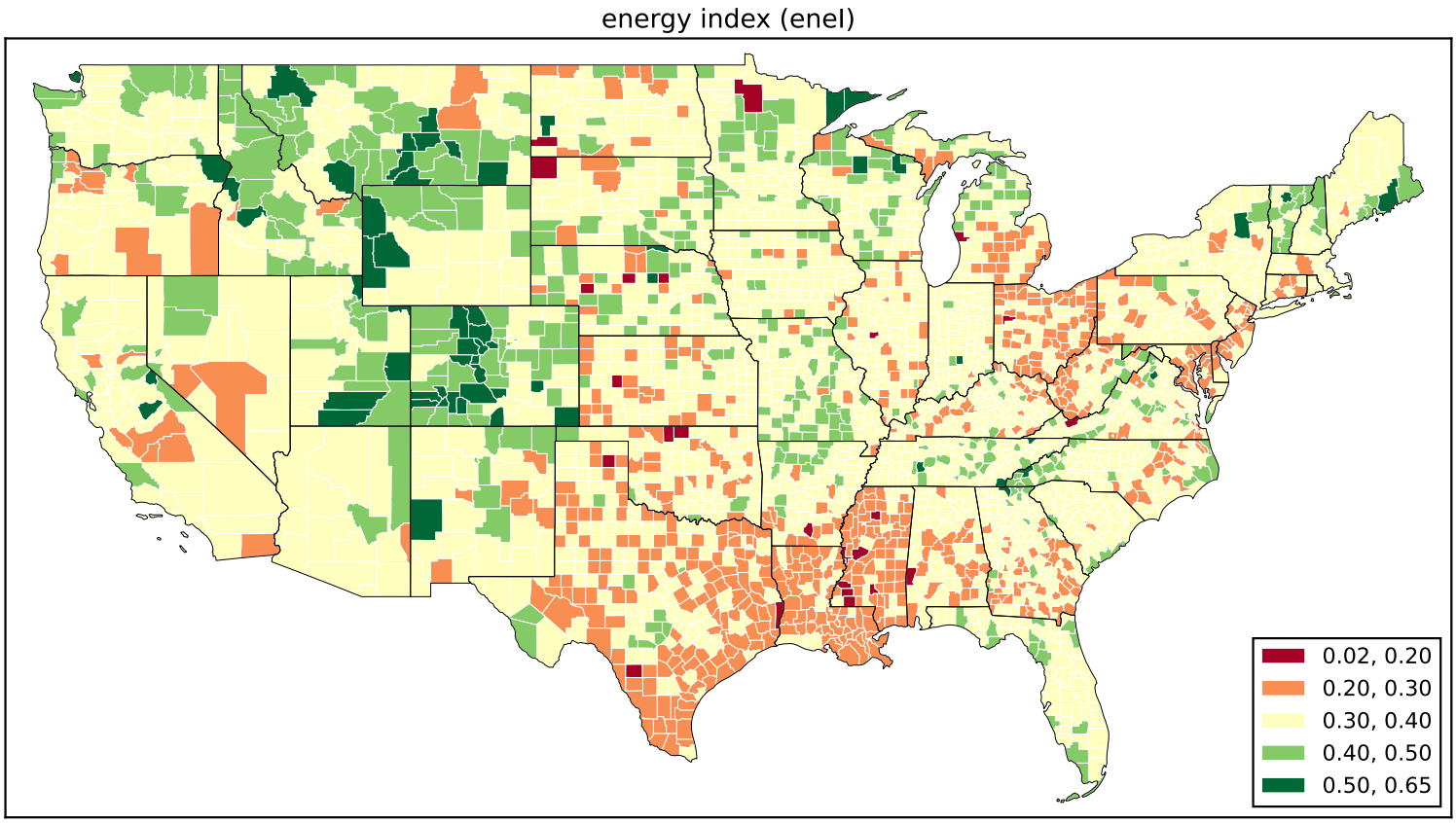


Figure 3: Energy/vitality index.

South does not surprise. As expected, the energy/vitality is low in South: especially in Mississippi and Louisiana, but also in most parts of Texas and many in Alabama and Georgia. Second comes Midwest[7], no surprise there. Most of Ohio is low on energy/vitality, and then much of Pennsylvania and West Virginia and South Michigan (North Michigan is high). Also much of Oklahoma and Kansas are low on energy/vitality. On West coast, perhaps surprisingly much of Oregon is low on energy/vitality.

As opposed to dull South and Midwest, there is definitely more energy/vitality in North East, for instance in New Jersey–one would think. But our results disagree–both Midwest and North East are low on energy/vitality, second only to South. Dull and low energy South does contrast with upbeat and high energy North East–it is just that the energy may not be that good in North East after all–stressed and neurotic. This concludes our general list of low energy/vitality states, but do note many red spots throughout the US and in other states. In appendix we have a top and bottom 100 ranking tables with all variables. Also in appendix are more detailed maps with more color bins. Overall they are very similar, so we stick with simplified exposition presented here. The extremes remain the same: dark red and dark green but there is much more gradation for mid-levels from light green through yellow to light red.

Turning now to high energy/vitality: in general West and North but not coastal/Pacific West: not Oregon, neither California. Colorado seems the highest energy/vitality, but also Utah, Wyoming, Montana, and Idaho.

[6]Here as maps show, it is only contiguous US, but in other analyses there were few additional observations from Alaska and Hawaii.
[7]And Northeast in next paragraph.

The happiness map in figure 4 broadly agrees with energy/vitality map in figure 3, and the correlation between the two is .83, for a scatterplot see appendix. There are few differences, however, for instance, Tennessee, Arkansas, and Missouri have more happiness than energy/vitality, North East, too, except New Jersey that is equally unhappy as it has low energy/vitality. As opposed to above energy/vitality index in the above map 3, the below happiness map in figure 4 is not entirely new.

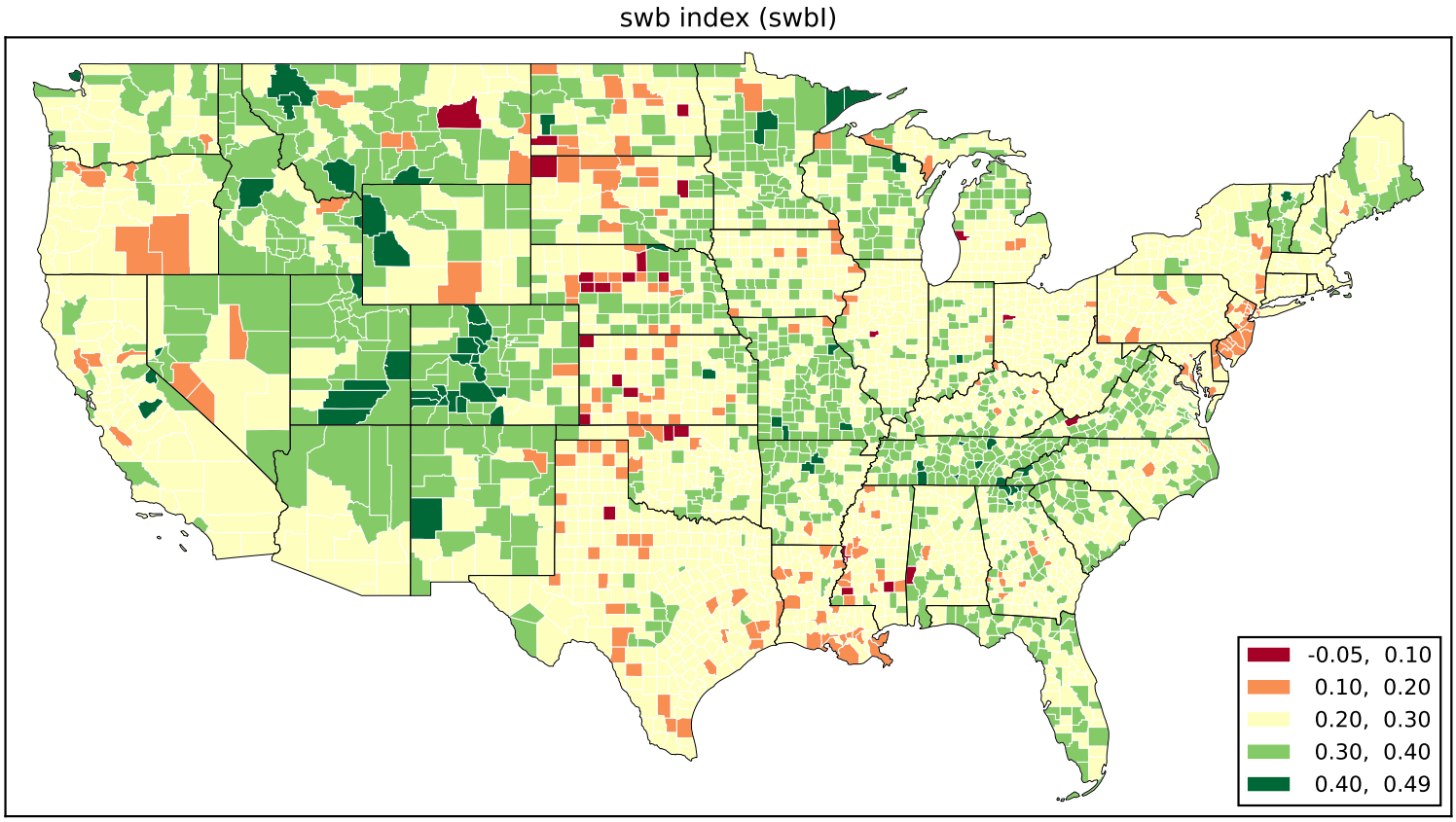


Figure 4: This figure was pulled directly from a web URL.

Results in figure 4 broadly agree at state level with Mitchell et al. (2013). Much of the results in our figure 4 also seem to agree at county level with Mitchell et al. (2013), but it is difficult to say as map averages at county level, while Mitchell et al. (2013) map closer to actual tweet locations and there are multiple points per county and it is difficult to estimate the average.[8]

Kjell et al. (2023) shows a map similar to our county estimates in figure 5.1C citing Jaidka et al. (2020). It looks like their sample of 1.5b is close to ours. But the results are different possibly due to their "interpolation of missing counties provided through a Gaussian process model using demographic and socioeconomic similarity between counties" (Kjell et al. 2023, fig. 5.1C caption): Alabama, Louisiana, and Mississippi are not unhappy, neither is North East, but Midwest is least happy (we find it unhappy but not as unhappy as Louisiana and Mississippi).

But maybe the most striking difference is that urban counties marked in their figure 5.1C with city names all appear to be the very happiest counties of all (dark green). Except Los Angeles and Miami (2nd shade of green, still very happy), Phoenix (yellow; average happiness), and per New York and New Orleans we cannot really tell as the black city dots cover polygons. Curiously, World Happiness Report and Gallup argue urban happiness despite data showing the contrary–see discussion in Okulicz-Kozaryn and Valente (2021). For instance see `https://news.gallup.com/opinion/gallup/315857/degree-urbanisation-effect-happiness.aspx`.

[8]Note that there is also an an Atlantic article that covers the arxiv version of their study.https://www.theatlantic.com/technology/archive/2013/02/the-geography-of-happiness-according-to-10-million-tweets/273286/

4.3 Zooming in on the largest cities

Table 5 shows wellbeing metrics for 20 largest counties. It is important to keep in mind that all data including population pertains to a county, not city, for instance Santa Ana is only about 300k but Orange county is about 3m. But we are using cities for ease of interpretations. First, summarizing all indicators as happy or unhappy as for most counties there is a clear pattern. First 4 numeric columns are energy/vitality (`eneI`, `posfunct`, `vitality`, `energy`) and subsequent 6 numeric columns are happiness (`swbI`, `lifesat`, `happiness`, `innerpeace`, `optimism`,`purpose`, and `hope`).

Happy counties are sunny places, but outside of unhappy Texas, and many coastal like Los Angeles, Phoenix, San Diego, Santa Ana, Miami, Las Vegas, San Jose and Ft Lauderdale. Also happy are cold but coastal Seattle and Manhattan (New York County).

Detroit, Queens, and Houston are the least happy. In addition to Houston, there are 2 other unhappy Texas cities, Dallas and Forth Worth. NYC has all happiness levels: Queens is very unhappy, Brooklyn average, and Manhattan very happy (it may be happy tourists tweeting happy tweets).

Then few interesting nuances. For instance 3 Texas cities Dallas, Forth Worth (neighbors), and San Antonio score poorly on energy/vitality (first 4 numeric columns) but quite well on hope. Queens and Brooklyn are lowest on hope.

At the bottom of the table is comparison to deciles from earlier figures (note that comparison colors in the table differ as they are scaled to current table). Decile 1 tends to be higher on first three indicators and lifesat, but lower on others.

Highest and lowest both energy/vitality and happiness indices tends to be in small counties, below median population size (see appendix). In appendix we also provide a full list of top and bottom counties on energy/vitality index `eneI`.

city	STNAME	CTYNAME	pop.(m)	eneI	posfunct	vitality	energy	swbI	lifesat	happiness	innerpeace	optimism	purpose	hope
Los Angeles	California	Los Angeles County	10.1	0.35	0.35	0.26	0.45	0.28	0.14	0.33	0.18	0.24	0.36	0.43
Chicago	Illinois	Cook County	5.2	0.34	0.33	0.23	0.45	0.26	0.10	0.32	0.15	0.23	0.36	0.42
Houston	Texas	Harris County	4.6	0.30	0.29	0.19	0.41	0.24	0.05	0.28	0.13	0.19	0.37	0.40
Phoenix	Arizona	Maricopa County	4.2	0.36	0.35	0.27	0.46	0.29	0.13	0.33	0.18	0.25	0.39	0.45
San Diego	California	San Diego County	3.3	0.38	0.37	0.30	0.46	0.30	0.16	0.35	0.19	0.25	0.38	0.44
Santa Ana	California	Orange County	3.1	0.38	0.37	0.29	0.47	0.30	0.17	0.35	0.20	0.26	0.38	0.45
Miami	Florida	Miami-Dade County	2.7	0.38	0.39	0.30	0.46	0.29	0.17	0.34	0.19	0.25	0.39	0.42
Brooklyn	New York	Kings County	2.6	0.34	0.33	0.24	0.45	0.22	0.07	0.29	0.14	0.16	0.34	0.34
Dallas	Texas	Dallas County	2.6	0.30	0.29	0.20	0.41	0.24	0.06	0.28	0.13	0.19	0.37	0.40
Riverside	California	Riverside County	2.3	0.32	0.31	0.22	0.42	0.25	0.11	0.30	0.16	0.19	0.32	0.41
Queens	New York	Queens County	2.3	0.31	0.29	0.20	0.43	0.19	0.02	0.25	0.11	0.12	0.32	0.30
Seattle	Washington	King County	2.1	0.38	0.37	0.29	0.48	0.29	0.13	0.34	0.17	0.25	0.40	0.45
San Bernardino	California	San Bernardino County	2.1	0.32	0.31	0.22	0.42	0.25	0.10	0.29	0.15	0.19	0.34	0.41
Las Vegas	Nevada	Clark County	2.1	0.39	0.38	0.30	0.49	0.30	0.18	0.36	0.19	0.27	0.39	0.44
Fort Worth	Texas	Tarrant County	2.0	0.29	0.28	0.19	0.41	0.23	0.06	0.27	0.13	0.18	0.36	0.39
San Jose	California	Santa Clara County	1.9	0.36	0.35	0.26	0.47	0.29	0.13	0.34	0.17	0.25	0.39	0.45
San Antonio	Texas	Bexar County	1.9	0.30	0.29	0.19	0.41	0.24	0.07	0.29	0.14	0.20	0.37	0.40
Fort Lauderdale	Florida	Broward County	1.9	0.37	0.37	0.29	0.45	0.28	0.14	0.32	0.18	0.23	0.40	0.42
Detroit	Michigan	Wayne County	1.8	0.27	0.27	0.15	0.40	0.21	0.02	0.26	0.10	0.17	0.32	0.39
Manhattan	New York	New York County	1.6	0.39	0.40	0.29	0.50	0.29	0.16	0.36	0.19	0.24	0.39	0.39
		decile 1 (<5.1k)		0.36	0.37	0.26	0.44	0.24	0.16	0.25	0.14	0.15	0.38	0.40
		mean of deciles 2-4 (5.1-18.5k)		0.35	0.34	0.25	0.45	0.28	0.13	0.31	0.18	0.23	0.40	0.44
		decile 10 (>204.9k-10077.3k)		0.33	0.32	0.23	0.44	0.26	0.09	0.30	0.15	0.21	0.37	0.42

Figure 5: Energy/vitality and happiness for 20 largest counties. Counties are sorted on population. At the bottom for comparison averages for 1st decile, 2-3rd deciles, and 10th decile.

Again, Kjell et al. (2023) has a map (fig 5.1C) similar to our county estimates, and the results differ. Their map has several large cities marked on it–and they appear to be the very happiest counties of all (most seem to be dark green). While the black dot marking

the city covers most or all of the city-counties, the metropolitan area bordering counties that are clearly visible are very happy, which already is surprising. Except Los Angeles and Miami (2nd shade of green, still very happy), and Phoenix (yellow; average happiness), and unclear New York and New Orleans (small counties completely covered by the black dot), it seems that all these large cities are very happy. They are from West to East: Seattle, Los Angeles, Salt Lake City, Phoenix, Denver, Minneapolis, Chicago, Houston, Atlanta, Washington, and Miami. For instance we have found Houston in table 5 to be unhappy, likewise earlier in county maps all of Houston vicinity wasn't happy.

Note that a caveat here as oposed to decile results earlier is that we have only a single county with limited number of tweets. For details refer to appendix "Descriptive Statistics" section.

5 Conclusion and discussion

This is the first study of geography of energy/vitality, where we find a very clear urban-rural gradient with energy/vitality raising from its lowest levels for large counties to highest level in small counties. Energy/vitality may be actually more important than happiness and at least as important, and it is striking that we know nothing about its geography.

It is important to clarify that HFGI indictors are about expressed wellbeing, not a survey-like measurement of well-being. These indicators capture expressions on Twitter/X.

References

Ballas, D. (2013): "What makes a 'happy city'?" *Cities*, 32, S39–S50.

Ballas, D. and R. Rijnks (2023): "Does Living on an Island Make You Happier?" in *International Conference on Computational Science and Its Applications*, 343–350.

Berry, B. J. and A. Okulicz-Kozaryn (2011): "An Urban-Rural Happiness Gradient," *Urban Geography*, 32, 871–883.

Cao, X., P. MacNaughton, Z. Deng, J. Yin, X. Zhang, and J. G. Allen (2018): "Using twitter to better understand the spatiotemporal patterns of public sentiment: A case study in Massachusetts, USA," *International journal of environmental research and public health*, 15, 250.

Carreno, D. F., N. Eisenbeck, J. A. Pérez-Escobar, and J. M. García-Montes (2021): "Inner harmony as an essential facet of well-being: A multinational study during the COVID-19 pandemic," *Frontiers in psychology*, 12, 648280.

Chai, Y., D. Kakkar, J. Palacios, and S. Zheng (2023): "Twitter sentiment geographical index dataset," *Scientific Data*, 10, 684.

Curini, L., S. Iacus, and L. Canova (2014): "Measuring Idiosyncratic Happiness Through the Analysis of Twitter: An Application to the Italian Case," *Social Indicators Research*, 1–18.

DePaulo, B. (2021): "Beyond Happiness: A Psychologically Rich Life Is a Good Life. What we get from psychological richness that happiness does not provide." *Psychology Today*.

Gupta, N. (2018): "Exploring happiness indicators in cities and industrial sectors using Twitter and Urban GIS data," Ph.D. thesis, University of Warwick.

Hanson, V. D. (2015): "The Oldest Divide. With roots dating back to our Founding, America's urban-rural split is wider than ever." *City Journal*, Autumn 2015.

Hollander, J. B. and H. Renski (2022): *Measuring urban attitudes using Twitter: An exploratory study*, Lincoln Institute of Land Policy.

Hu, T., B. She, L. Duan, H. Yue, and J. Clunis (2019): "A systematic spatial and temporal sentiment analysis on geo-tweets," *Ieee Access*, 8, 8658–8667.

Iacus, S. M., D. Jain, A. Nasuto, G. Porro, M. Carammia, and A. Vezzulli (2025): "The Human Flourishing Geographic Index: A County-Level Dataset for the United States, 2013–2023," *arXiv preprint arXiv:2511.03915*.

Iacus, S. M. and G. Porro (2025): "Two Americas of Well-Being: Divergent Rural-Urban Patterns of Life Satisfaction and Happiness from 2.6 B Social Media Posts," .

Iacus, S. M., G. Porro, S. Salini, and E. Siletti (2015): "Social networks, happiness and health: from sentiment analysis to a multidimensional indicator of subjective well-being," *arXiv preprint arXiv:1512.01569*.

Jaidka, K., S. Giorgi, H. A. Schwartz, M. L. Kern, L. H. Ungar, and J. C. Eichstaedt (2020): "Estimating geographic subjective well-being from Twitter: A comparison of dictionary and data-driven language methods," *Proceedings of the national academy of sciences*, 117, 10165–10171.

Jaidka, K., S. C. Guntuku, J. H. Lee, Z. Luo, A. Buffone, and L. H. Ungar (2021): "The rural–urban stress divide: Obtaining geographical insights through Twitter," *Computers in Human Behavior*, 114, 106544.

Jensen, E. A. (2017): "Putting the methodological brakes on claims to measure national happiness through Twitter: Methodological limitations in social media analytics," *PloS one*, 12, e0180080.

Kalaidis, J. (2014): "Why I Miss the Suburbs," *Atlantic Cities*.

Karamouzas, D., I. Mademlis, and I. Pitas (2022): "Public opinion monitoring through collective semantic analysis of tweets," *Social Network Analysis and Mining*, 12, 91.

Kjell, O., S. Giorgi, H. A. Schwartz, and J. C. Eichstaedt (2023): "Towards well-being measurement with social media across space, time and cultures: Three generations of progress," *World happiness report 2023*, 131–162.

Knies, G. (2026): "Beyond the Rural Happiness Paradox: Welfare Regimes, Settlement Types, and Multidimensional Subjective Wellbeing in Europe," *Social Indicators Research*, 183, 33.

Leetaru, K., S. Wang, G. Cao, A. Padmanabhan, and E. Shook (2013): "Mapping the global Twitter heartbeat: The geography of Twitter," *First Monday*.

Lenzi, C. and G. Perucca (2016a): "Are urbanized areas source of life satisfaction? Evidence from EU regions," *Papers in Regional Science*.

——— (2016b): "The Easterlin paradox and the urban-rural divide in life satisfaction: Evidence from Romania," *Unpublished; http://www.grupposervizioambiente.it*.

——— (2016c): "Life Satisfaction across Cities: Evidence from Romania," *The Journal of Development Studies*, 1–16.

——— (2020): "Not too close, not too far: Urbanisation and life satisfaction along the urban hierarchy," *Urban Studies*, 0042098020962397.

——— (2022): "No Place for Poor Men: On the Asymmetric Effect of Urbanization on Life Satisfaction," *Social Indicators Research*, 1–23.

Malik, M., H. Lamba, C. Nakos, and J. Pfeffer (2015): "Population bias in geotagged tweets," in *proceedings of the international AAAI conference on web and social media*, vol. 9, 18–27.

Manson, M. (2015): *The subtle art of not giving a fuck*, Harper One.

——— (2019): *Everything Is F*cked: A Book About Hope*.

Michalos, A. C. (1980): "Satisfaction and happiness," *Social indicators research*, 8, 385–422.

Mitchell, L., M. R. Frank, K. D. Harris, P. S. Dodds, and C. M. Danforth (2013): "The geography of happiness: Connecting twitter sentiment and expression, demographics, and objective characteristics of place," *PloS one*, 8, e64417.

Nguyen, Q. C., S. Kath, H.-W. Meng, D. Li, K. R. Smith, J. A. VanDerslice, M. Wen, and F. Li (2016): "Leveraging geotagged Twitter data to examine neighborhood happiness, diet, and physical activity," *Applied Geography*, 73, 77–88.

Okulicz-Kozaryn, A. (2023): "Unhappy metros: Panel evidence," *Applied Research in Quality of Life*, 18, 753–763.

——— (2024): "Urban regrets (Unhappy metros: Satisfaction with life scale (SWLS))," *Heliyon*, 10.

Okulicz-Kozaryn, A., B. K. Everett, and E. Mikhaeil (2024): "Happiness is In The air if it grows growing places are happier than shrinking ones," *Urban Affairs Review*, 60, 1594–1613.

Okulicz-Kozaryn, A. and J. M. Mazelis (2018): "Urbanism and Happiness: A Test of Wirth's Theory on Urban Life," *Urban Studies*, 55, 349–364.

Okulicz-Kozaryn, A. and R. Valente (2020): "The perennial dissatisfaction of urban upbringing," *Cities*, 104, 102751.

Okulicz-Kozaryn, A. and R. R. Valente (2021): "Urban unhappiness is common," *Cities*, 118, 103368.

Park, Y., M. Kim, and K. Seong (2021): "Happy neighborhoods: Investigating neighborhood conditions and sentiments of a shrinking city with Twitter data," *Growth and change*, 52, 539–566.

Pleeging, E., M. Burger, and J. van Exel (2021): "The relations between hope and subjective well-being: A literature overview and empirical analysis," *Applied Research in Quality of Life*, 16, 1019–1041.

Plunz, R. A., Y. Zhou, M. I. C. Vintimilla, K. Mckeown, T. Yu, L. Uguccioni, and M. P. Sutto (2019): "Twitter sentiment in New York City parks as measure of well-being," *Landscape and urban planning*, 189, 235–246.

Ryff, C. D. (1989): "Happiness is everything, or is it? Explorations on the meaning of psychological well-being." *Journal of personality and social psychology*, 57, 1069.

——— (2014): "Psychological well-being revisited: Advances in the science and practice of eudaimonia," *Psychotherapy and psychosomatics*, 83, 10–28.

Ryff, C. D. and B. H. Singer (2008): "Know thyself and become what you are: A eudaimonic approach to psychological well-being," *Journal of Happiness Studies*, 9, 13–39.

Shah, M., J. Javed, W. Iqbal, and S. A. Bibi (2026): "The Impact of Optimism on Life Satisfaction, Psychological Well-being, and Happiness: The Mediating Role of Resilience," *Policy Journal of Social Science Review*, 4, 857–871.

Song, M. and N. Zhao (2023): "Predicting life satisfaction based on the emotion words in self-statement texts," *Frontiers in psychiatry*, 14, 1121915.

Sørensen, J. F. (2014): "Rural–urban differences in life satisfaction: Evidence from the European Union," *Regional Studies*, 48, 1451–1466.

Sørensen, J. F. L. (2024): "Why are City Residents Less Happy than the Rest of the Population in Developed Countries? Studying the Urban-Rural Happiness Gap in Denmark Using Blinder-Oaxaca Decomposition," *Applied Research in Quality of Life*, 1–20.

Tan, M. J. and C. Guan (2021): "Are people happier in locations of high property value? Spatial temporal analytics of activity frequency, public sentiment and housing price using twitter data," *Applied Geography*, 132, 102474.

Weckroth, M., S. Ala-Mantila, D. Ballas, T. Ziogas, and J. Ikonen (2022): "Urbanity, Neighbourhood Characteristics and Perceived Quality of Life (QoL): Analysis of Individual and Contextual Determinants for Perceived QoL in 3300 Postal Code Areas in Finland," *Social Indicators Research*, 1–26.

Winters, J. V., Y. Li, et al. (2015): "Urbanization, Natural Amenities, and Subjective Well-Being: Evidence from US Counties," Tech. rep., Institute for the Study of Labor (IZA).

Zika, S. and K. Chamberlain (1992): "On the relation between meaning in life and psychological well-being," *British Journal of Psychology*, 83, 133–145.

Ziogas, T., D. Ballas, S. Koster, and A. Edzes (2020): "How happy are my neighbours? Modelling spatial spillover effects of well-being," *arXiv preprint arXiv:2007.11580*.

——— (2023): "Happiness, Space and Place: Community Area Clustering and Spillovers of Life Satisfaction in Canada," *Applied Research in Quality of Life*, 1–44.

ONLINE APPENDIX

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead]

5.1 Descriptive Statistics

Full descriptive statistics including each indicator's value, standar error, salience, number of tweets, and number of valid tweets by county and state is `https://raw.githubusercontent.com/theaok/hfgi/refs/heads/main/desSta.csv`.

	count	mean	std	min	25%	50%	75%	max
enel	3142.0	0.34	0.06	0.02	0.30	0.34	0.38	0.65
posfunct	3142.0	0.34	0.07	-0.18	0.30	0.34	0.38	0.64
vitality	3142.0	0.24	0.09	-0.17	0.18	0.23	0.29	0.70
energy	3142.0	0.45	0.05	0.13	0.42	0.45	0.48	0.64
swbl	3142.0	0.28	0.06	-0.13	0.24	0.28	0.31	0.49
lifesat	3142.0	0.12	0.08	-0.39	0.08	0.12	0.16	0.43
happiness	3142.0	0.30	0.06	-0.19	0.28	0.31	0.34	0.50
innerpeace	3142.0	0.17	0.06	-0.24	0.14	0.17	0.20	0.43
optimism	3142.0	0.22	0.08	-0.35	0.19	0.23	0.27	0.44
purpose	3141.0	0.40	0.07	-0.17	0.36	0.40	0.44	0.69
hope	3142.0	0.44	0.06	0.02	0.41	0.44	0.48	0.62

Figure 6: Short descriptive statistics.

5.2 Additional results

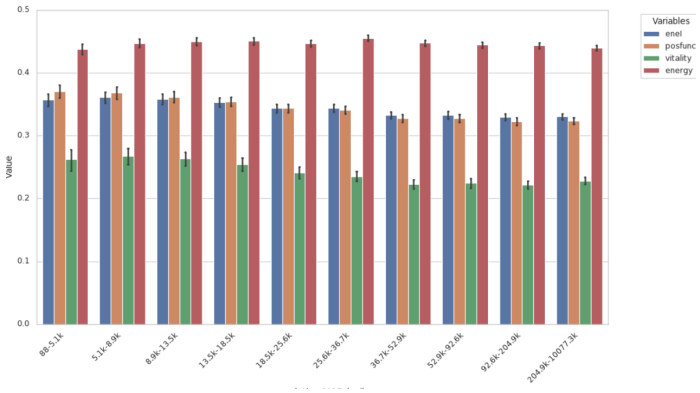


Figure 7: Energy/vitality index and its dimensions.

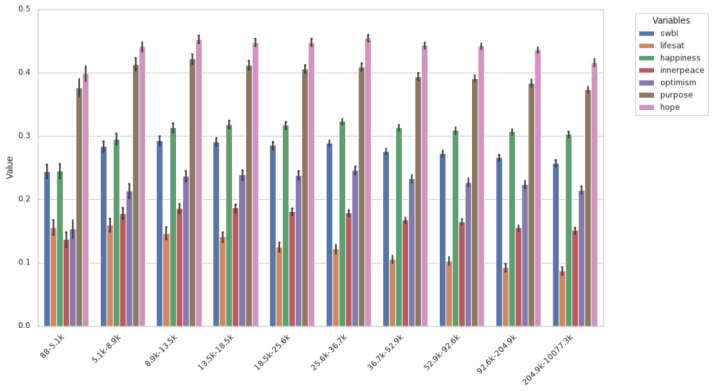


Figure 8: Happiness index and its dimensions.

5.3 Additional results (without dropping <=2012)

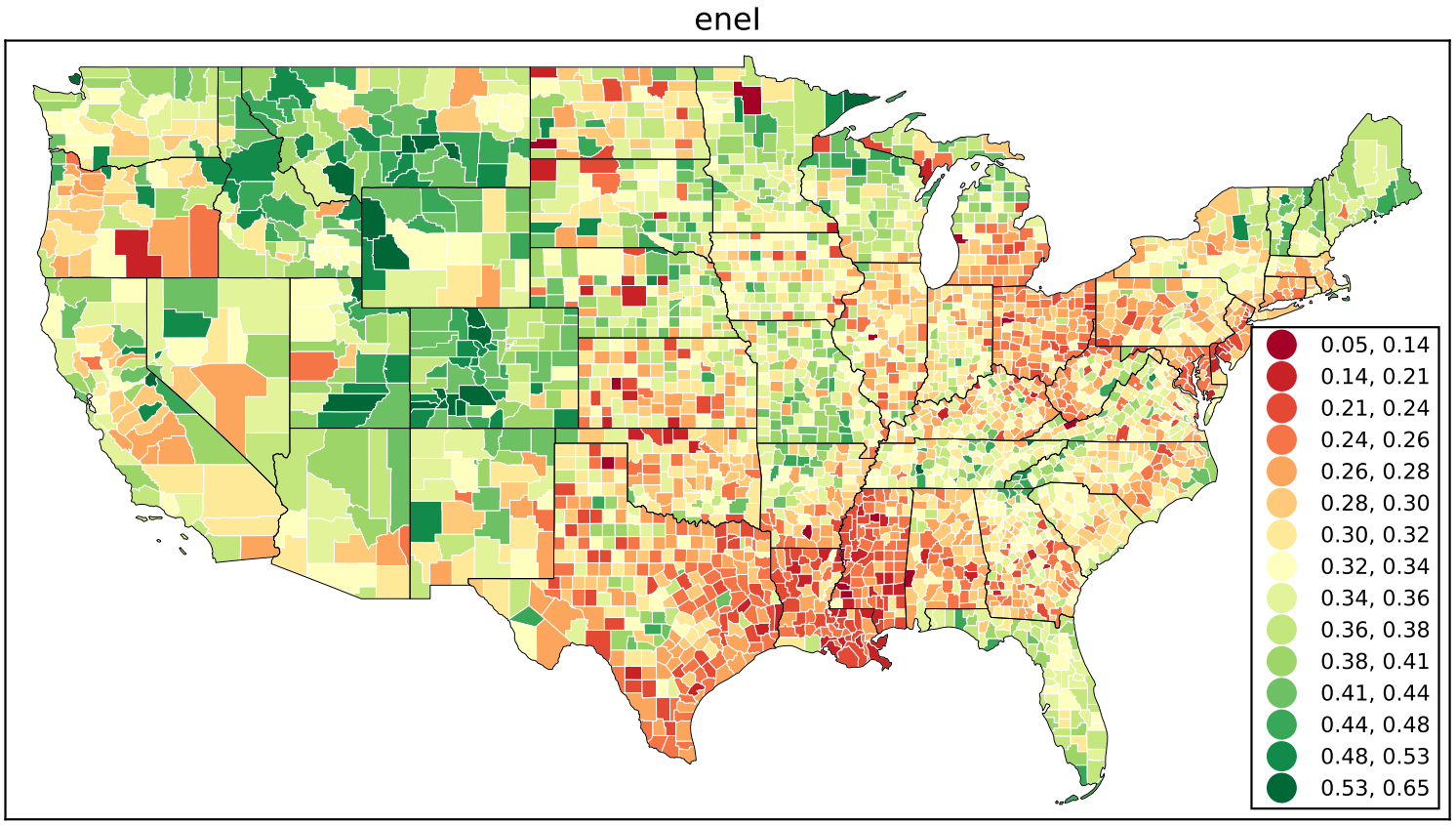


Figure 9: more classes This figure was pulled directly from a web URL.

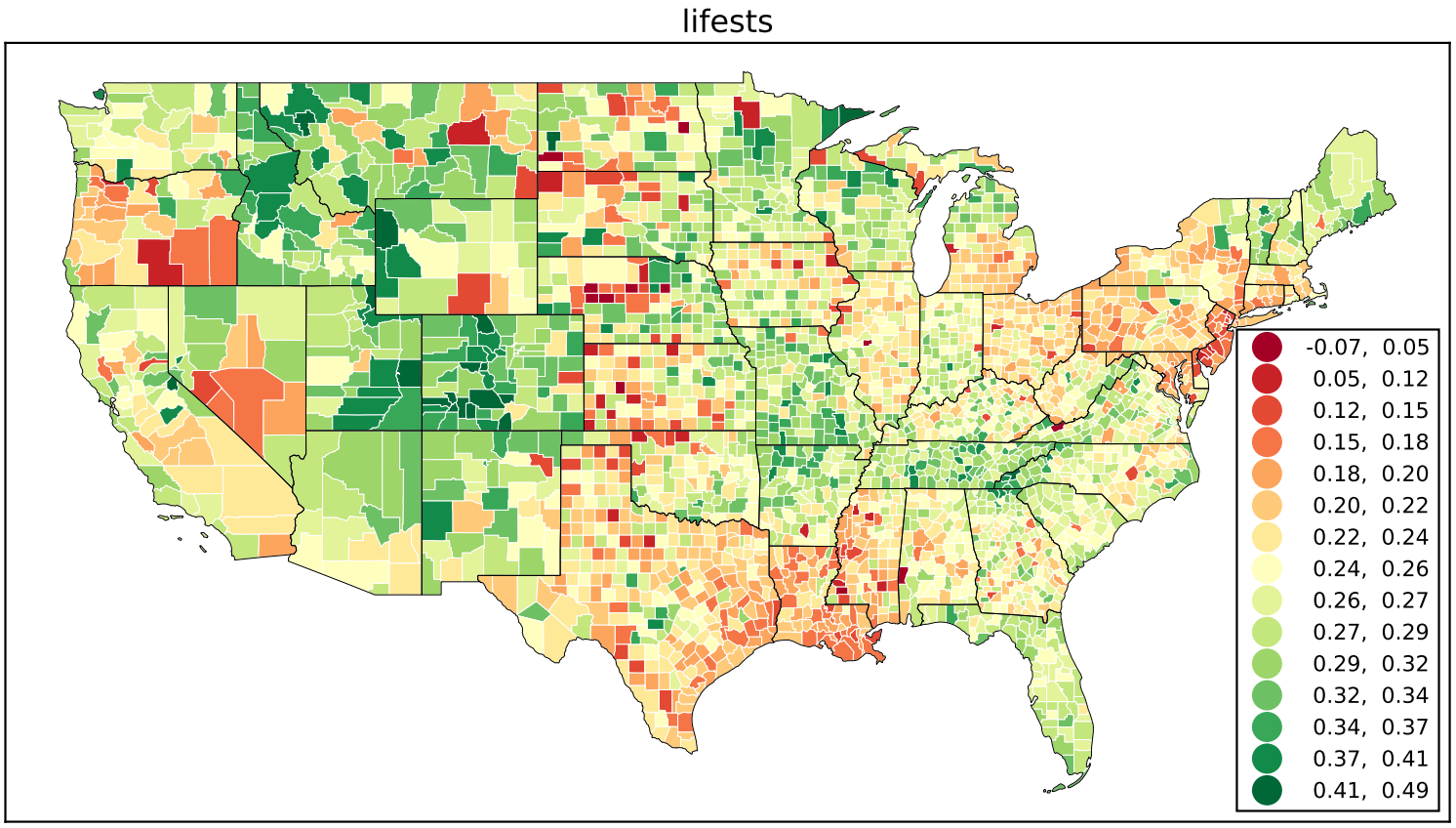


Figure 10: more classes This figure was pulled directly from a web URL.

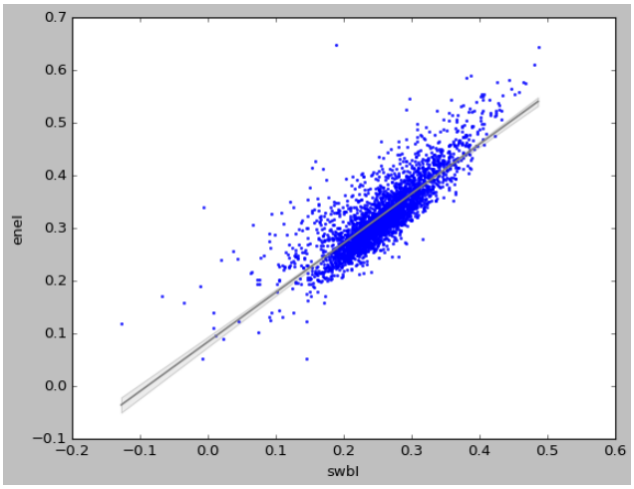


Figure 11: Correlation is 0.83.

STNAME	CTYNAME	POPESTIMATE2015	enel	posfunct	vitality	energy	swbl	lifesat	happiness	innerpeace	optimism	purpose	hope
Montana	Golden Valley County	818	0.65	0.64	0.70	0.60	0.19	-0.02	0.19	-0.11	0.11	0.50	0.47
Colorado	Jackson County	1352	0.64	0.61	0.68	0.64	0.49	0.43	0.48	0.39	0.41	0.64	0.59
Colorado	Hinsdale County	762	0.61	0.62	0.65	0.56	0.48	0.40	0.50	0.40	0.44	0.54	0.61
Montana	Musselshell County	4631	0.59	0.59	0.57	0.61	0.39	0.24	0.39	0.28	0.33	0.56	0.52
Wyoming	Sublette County	10064	0.59	0.57	0.61	0.58	0.38	0.19	0.33	0.31	0.27	0.64	0.56
Alaska	Skagway Municipality	1090	0.58	0.57	0.62	0.55	0.45	0.40	0.41	0.33	0.34	0.59	0.60
Wyoming	Teton County	23083	0.58	0.55	0.60	0.59	0.47	0.38	0.43	0.32	0.40	0.64	0.62
California	Alpine County	1080	0.58	0.57	0.61	0.55	0.43	0.34	0.40	0.33	0.30	0.58	0.61
Utah	Rich County	2301	0.58	0.57	0.59	0.56	0.47	0.42	0.49	0.35	0.44	0.56	0.55
Colorado	Saguache County	6250	0.56	0.57	0.53	0.59	0.45	0.35	0.45	0.38	0.40	0.53	0.58
Colorado	Ouray County	4611	0.56	0.53	0.56	0.59	0.46	0.38	0.46	0.33	0.41	0.56	0.62
Colorado	Dolores County	1980	0.55	0.59	0.57	0.50	0.41	0.38	0.42	0.27	0.34	0.53	0.50
Colorado	San Juan County	690	0.55	0.57	0.57	0.53	0.45	0.41	0.49	0.30	0.38	0.54	0.59
Utah	Garfield County	4966	0.55	0.56	0.57	0.53	0.40	0.28	0.41	0.35	0.32	0.58	0.48
Minnesota	Cook County	5249	0.55	0.51	0.57	0.57	0.44	0.36	0.44	0.35	0.35	0.54	0.56
Montana	Madison County	8033	0.55	0.55	0.54	0.57	0.40	0.35	0.38	0.31	0.26	0.54	0.57
Washington	San Juan County	16155	0.55	0.53	0.54	0.58	0.41	0.30	0.43	0.32	0.36	0.50	0.53
Montana	Treasure County	680	0.55	0.57	0.53	0.54	0.30	0.24	0.31	0.17	0.25	0.34	0.48
Tennessee	Hancock County	6577	0.54	0.51	0.53	0.59	0.36	0.24	0.40	0.19	0.37	0.44	0.52
Colorado	Mineral County	742	0.54	0.55	0.58	0.50	0.42	0.38	0.40	0.29	0.35	0.53	0.54
Colorado	Grand County	14742	0.54	0.51	0.56	0.56	0.41	0.28	0.42	0.25	0.36	0.55	0.58
Tennessee	Perry County	7855	0.54	0.61	0.52	0.49	0.40	0.35	0.34	0.29	0.32	0.62	0.46
Hawaii	Kauai County	71052	0.54	0.51	0.56	0.54	0.43	0.32	0.45	0.32	0.38	0.52	0.57
Hawaii	Maui County	163993	0.53	0.51	0.55	0.54	0.43	0.31	0.45	0.31	0.38	0.57	0.56
Utah	Daggett County	1107	0.53	0.50	0.59	0.51	0.38	0.34	0.37	0.35	0.35	0.39	0.47
Wyoming	Lincoln County	18788	0.53	0.52	0.52	0.55	0.37	0.25	0.33	0.23	0.25	0.58	0.60
Idaho	Boise County	6963	0.53	0.51	0.53	0.53	0.39	0.27	0.40	0.27	0.32	0.52	0.53
Colorado	Custer County	4434	0.53	0.53	0.49	0.56	0.43	0.34	0.44	0.35	0.36	0.51	0.58
Montana	Powder River County	1768	0.52	0.53	0.55	0.49	0.29	0.16	0.26	0.12	0.18	0.50	0.54
Colorado	San Miguel County	7850	0.52	0.51	0.52	0.54	0.43	0.35	0.46	0.27	0.38	0.57	0.57
Utah	Kane County	7045	0.52	0.51	0.53	0.53	0.41	0.32	0.40	0.32	0.33	0.53	0.54
California	Mariposa County	17653	0.52	0.51	0.54	0.51	0.39	0.35	0.41	0.30	0.34	0.50	0.47
Utah	Grand County	9530	0.52	0.52	0.53	0.51	0.43	0.35	0.42	0.31	0.37	0.55	0.55
Montana	Stillwater County	9419	0.52	0.51	0.47	0.58	0.34	0.21	0.35	0.20	0.24	0.48	0.54
Minnesota	Clearwater County	8794	0.52	0.53	0.46	0.57	0.39	0.28	0.39	0.31	0.36	0.50	0.51
Wisconsin	Price County	13577	0.52	0.50	0.50	0.56	0.38	0.25	0.38	0.30	0.35	0.47	0.56
Colorado	Costilla County	3576	0.52	0.50	0.49	0.57	0.43	0.34	0.44	0.33	0.39	0.49	0.56
Alaska	Haines Borough	2519	0.52	0.50	0.57	0.48	0.36	0.33	0.31	0.28	0.16	0.55	0.51
Minnesota	Lake County	10540	0.52	0.51	0.51	0.54	0.40	0.37	0.39	0.26	0.33	0.53	0.53
Colorado	Lake County	7455	0.52	0.49	0.53	0.54	0.42	0.29	0.42	0.30	0.36	0.57	0.56
Colorado	Pitkin County	17962	0.52	0.49	0.49	0.57	0.42	0.32	0.46	0.28	0.38	0.50	0.58
Montana	Petroleum County	475	0.52	0.56	0.45	0.53	0.37	0.19	0.36	0.39	0.20	0.54	0.53
Colorado	Crowley County	5633	0.51	0.50	0.46	0.58	0.36	0.30	0.33	0.25	0.23	0.65	0.40
North Dakota	Billings County	950	0.51	0.50	0.50	0.54	0.42	0.34	0.38	0.32	0.34	0.58	0.58
North Carolina	Swain County	14325	0.51	0.51	0.52	0.50	0.42	0.29	0.42	0.33	0.36	0.57	0.57
New Mexico	Catron County	3480	0.51	0.48	0.50	0.55	0.40	0.40	0.39	0.26	0.31	0.52	0.54
Nebraska	Boyd County	2004	0.51	0.55	0.46	0.52	0.43	0.35	0.43	0.34	0.36	0.57	0.55
Idaho	Adams County	3891	0.51	0.49	0.51	0.52	0.37	0.25	0.39	0.20	0.31	0.47	0.57
Colorado	Baca County	3554	0.51	0.48	0.54	0.50	0.35	0.28	0.28	0.29	0.18	0.54	0.51
Nebraska	Garfield County	2013	0.50	0.49	0.50	0.53	0.38	0.27	0.32	0.32	0.30	0.53	0.55
Virginia	Rappahannock County	7409	0.50	0.51	0.44	0.56	0.40	0.27	0.45	0.28	0.36	0.51	0.54
New York	Hamilton County	4700	0.50	0.45	0.50	0.56	0.36	0.18	0.39	0.26	0.29	0.54	0.47
Hawaii	Hawaii County	196111	0.50	0.49	0.52	0.50	0.37	0.30	0.36	0.29	0.21	0.51	0.52
Montana	Flathead County	95940	0.50	0.50	0.45	0.55	0.41	0.27	0.42	0.27	0.37	0.56	0.55
Montana	Meagher County	1822	0.50	0.52	0.46	0.51	0.31	0.22	0.33	0.18	0.22	0.41	0.49
Nevada	Storey County	3881	0.50	0.48	0.43	0.58	0.40	0.27	0.42	0.26	0.36	0.55	0.56
Washington	Skamania County	11386	0.50	0.50	0.48	0.50	0.38	0.29	0.40	0.29	0.29	0.50	0.51
South Dakota	Jackson County	3281	0.49	0.50	0.48	0.50	0.37	0.30	0.39	0.25	0.30	0.49	0.51
Colorado	Summit County	30064	0.49	0.46	0.49	0.53	0.39	0.27	0.43	0.25	0.35	0.51	0.53
South Dakota	Custer County	8456	0.49	0.47	0.48	0.53	0.39	0.35	0.38	0.27	0.30	0.52	0.51
Idaho	Idaho County	16312	0.49	0.50	0.45	0.53	0.39	0.29	0.39	0.28	0.31	0.53	0.54
California	Sierra County	2990	0.49	0.51	0.49	0.48	0.32	0.32	0.24	0.31	0.06	0.48	0.52
Colorado	Clear Creek County	9246	0.49	0.51	0.55	0.41	0.37	0.35	0.35	0.14	0.27	0.58	0.56
Oregon	Wallowa County	6810	0.49	0.56	0.55	0.36	0.37	0.30	0.33	0.31	0.25	0.57	0.42
Utah	Wayne County	2689	0.49	0.49	0.48	0.49	0.41	0.35	0.41	0.32	0.32	0.49	0.55
Vermont	Lamoille County	25265	0.49	0.48	0.46	0.53	0.40	0.26	0.42	0.27	0.35	0.55	0.56
Nevada	Pershing County	6620	0.49	0.48	0.43	0.56	0.31	0.14	0.34	0.14	0.22	0.52	0.50
Montana	Carbon County	10404	0.49	0.46	0.48	0.52	0.35	0.23	0.35	0.17	0.23	0.54	0.58
Oregon	Hood River County	22925	0.49	0.51	0.50	0.45	0.35	0.21	0.39	0.25	0.29	0.46	0.51
Oregon	Wheeler County	1327	0.48	0.57	0.52	0.36	0.37	0.32	0.37	0.30	0.23	0.53	0.46
Montana	Granite County	3160	0.48	0.51	0.46	0.49	0.37	0.29	0.36	0.24	0.31	0.54	0.51
Montana	Jefferson County	11576	0.48	0.47	0.45	0.53	0.33	0.23	0.33	0.24	0.23	0.44	0.49
Colorado	Eagle County	53863	0.48	0.46	0.45	0.55	0.40	0.28	0.44	0.25	0.37	0.49	0.56
Colorado	Montezuma County	25763	0.48	0.49	0.43	0.53	0.37	0.24	0.38	0.25	0.34	0.48	0.51
Idaho	Fremont County	12824	0.48	0.48	0.44	0.53	0.34	0.23	0.36	0.22	0.27	0.47	0.51
Missouri	Hickory County	9269	0.48	0.49	0.45	0.50	0.40	0.28	0.42	0.31	0.33	0.52	0.52
Kansas	Morris County	5630	0.48	0.48	0.38	0.57	0.41	0.32	0.35	0.33	0.34	0.55	0.58
Michigan	Keweenaw County	2124	0.48	0.48	0.45	0.50	0.33	0.28	0.34	0.22	0.25	0.42	0.49
South Dakota	Marshall County	4830	0.48	0.52	0.39	0.53	0.39	0.30	0.35	0.25	0.30	0.54	0.57
Colorado	Chaffee County	18635	0.48	0.49	0.44	0.50	0.38	0.24	0.40	0.25	0.35	0.50	0.54
Wisconsin	Buffalo County	13245	0.48	0.46	0.44	0.54	0.38	0.32	0.34	0.31	0.26	0.51	0.54
South Dakota	Gregory County	4195	0.48	0.49	0.46	0.48	0.31	0.29	0.26	0.28	0.16	0.44	0.47
Montana	Sweet Grass County	3618	0.48	0.46	0.43	0.54	0.32	0.30	0.30	0.14	0.22	0.45	0.49
Montana	Lake County	29482	0.47	0.48	0.38	0.57	0.42	0.22	0.47	0.25	0.42	0.58	0.59
Colorado	Conejos County	8064	0.47	0.51	0.45	0.46	0.38	0.31	0.41	0.30	0.30	0.45	0.52
Tennessee	Van Buren County	5697	0.47	0.49	0.47	0.46	0.38	0.24	0.38	0.29	0.32	0.53	0.52
Indiana	Brown County	15003	0.47	0.43	0.46	0.52	0.38	0.28	0.40	0.20	0.34	0.52	0.52
Colorado	Sedgwick County	2373	0.47	0.46	0.45	0.51	0.31	0.19	0.34	0.21	0.22	0.43	0.47
Montana	Toole County	5083	0.47	0.46	0.36	0.59	0.38	0.21	0.41	0.20	0.37	0.52	0.54
North Dakota	Oliver County	1866	0.47	0.47	0.47	0.48	0.32	0.07	0.40	0.10	0.36	0.39	0.59
Tennessee	Fentress County	17974	0.47	0.49	0.38	0.53	0.40	0.30	0.30	0.29	0.37	0.58	0.58
South Dakota	Brule County	5239	0.47	0.46	0.44	0.50	0.32	0.18	0.32	0.20	0.24	0.48	0.49
Hawaii	Honolulu County	991755	0.47	0.44	0.46	0.50	0.36	0.23	0.40	0.25	0.32	0.47	0.51
North Carolina	Avery County	17494	0.47	0.45	0.47	0.49	0.38	0.29	0.39	0.25	0.33	0.49	0.55
Wyoming	Niobrara County	2504	0.47	0.45	0.49	0.47	0.29	0.21	0.24	0.17	0.17	0.43	0.52
Alaska	Ketchikan Gateway Borough	13737	0.47	0.45	0.45	0.50	0.37	0.24	0.37	0.25	0.29	0.53	0.53
Maine	Hancock County	54280	0.47	0.44	0.45	0.51	0.37	0.28	0.39	0.24	0.34	0.45	0.53
Montana	Judith Basin County	1932	0.47	0.47	0.38	0.55	0.39	0.23	0.39	0.31	0.35	0.48	0.58
Montana	Fallon County	3172	0.47	0.45	0.40	0.54	0.28	0.28	0.16	0.17	0.15	0.46	0.47
Alaska	Bristol Bay Borough	906	0.47	0.44	0.50	0.46	0.31	0.18	0.32	0.20	0.22	0.46	0.48

Table 1: Top 100 Sorted on `eneI`.

STNAME	CTYNAME	POPESTIMATE2015	enel	posfunct	vitality	energy	swbl	lifesat	happiness	innerpeace	optimism	purpose	hope
Texas	Jasper County	35268	0.22	0.20	0.12	0.34	0.18	0.03	0.19	0.12	0.09	0.29	0.36
Louisiana	Tangipahoa Parish	128774	0.22	0.20	0.11	0.35	0.18	-0.01	0.20	0.09	0.12	0.32	0.35
Texas	Liberty County	79537	0.22	0.21	0.10	0.35	0.16	-0.01	0.19	0.07	0.09	0.27	0.34
Michigan	Gogebic County	15537	0.22	0.19	0.10	0.37	0.15	-0.04	0.21	0.06	0.12	0.23	0.33
North Dakota	Sioux County	4385	0.22	0.18	0.06	0.41	0.17	-0.03	0.22	0.05	0.14	0.29	0.36
West Virginia	Brooke County	23213	0.22	0.22	0.09	0.35	0.16	-0.01	0.21	0.05	0.09	0.28	0.36
Louisiana	St. Landry Parish	83693	0.22	0.20	0.09	0.36	0.18	-0.02	0.21	0.10	0.13	0.29	0.36
Georgia	Jefferson County	15926	0.22	0.22	0.09	0.34	0.20	0.03	0.23	0.09	0.16	0.29	0.39
Louisiana	De Soto Parish	27112	0.22	0.18	0.10	0.37	0.17	-0.06	0.19	0.05	0.12	0.36	0.36
Louisiana	Terrebonne Parish	113894	0.22	0.19	0.11	0.35	0.17	-0.02	0.20	0.07	0.12	0.29	0.36
Texas	Dimmit County	10914	0.22	0.21	0.10	0.33	0.18	0.01	0.22	0.10	0.12	0.27	0.37
Louisiana	St. James Parish	21483	0.22	0.20	0.11	0.34	0.17	-0.02	0.19	0.09	0.11	0.31	0.33
West Virginia	Tyler County	8959	0.22	0.29	-0.00	0.36	0.24	0.07	0.26	0.13	0.17	0.39	0.44
Alaska	Yukon-Koyukuk Census Area	5461	0.22	0.25	0.19	0.22	0.07	0.07	-0.08	0.09	-0.27	0.25	0.35
Oklahoma	Woods County	9306	0.22	0.24	0.07	0.33	0.17	0.04	0.16	0.11	0.02	0.27	0.42
Texas	Brooks County	7203	0.22	0.22	0.13	0.29	0.16	0.02	0.19	0.02	0.11	0.31	0.33
Louisiana	St. John the Baptist Parish	43527	0.22	0.20	0.12	0.33	0.16	-0.04	0.18	0.09	0.09	0.29	0.35
Alaska	Northwest Arctic Borough	7787	0.22	0.19	0.17	0.28	0.06	0.05	0.03	-0.01	-0.17	0.20	0.28
Louisiana	Madison Parish	11560	0.21	0.19	0.12	0.34	0.17	-0.02	0.19	0.09	0.11	0.29	0.34
Georgia	Clinch County	6828	0.21	0.18	0.06	0.40	0.23	0.08	0.24	0.12	0.18	0.35	0.41
Michigan	Shiawassee County	68538	0.21	0.21	0.08	0.36	0.16	-0.01	0.23	0.03	0.13	0.24	0.34
Nebraska	Keya Paha County	797	0.21	0.21	0.11	0.33	0.17	0.13	0.22	-0.01	-0.05	0.39	0.33
Indiana	Wayne County	67044	0.21	0.16	0.09	0.39	0.15	-0.09	0.21	0.02	0.11	0.23	0.40
Texas	Duval County	11313	0.21	0.21	0.13	0.30	0.15	-0.06	0.20	0.06	0.11	0.25	0.36
Texas	Scurry County	17565	0.21	0.21	0.08	0.35	0.20	0.03	0.23	0.09	0.13	0.30	0.41
Georgia	Calhoun County	6536	0.21	0.31	0.01	0.32	0.22	0.05	0.23	0.09	0.15	0.40	0.43
Louisiana	Bienville Parish	13828	0.21	0.20	0.12	0.32	0.16	-0.03	0.18	0.08	0.09	0.28	0.35
Mississippi	Claiborne County	9211	0.21	0.17	0.11	0.35	0.17	-0.02	0.19	0.05	0.11	0.34	0.35
Texas	Hardin County	55767	0.21	0.19	0.08	0.36	0.16	0.04	0.19	0.08	0.08	0.22	0.37
Mississippi	Pearl River County	54978	0.21	0.20	0.06	0.37	0.19	-0.02	0.23	0.09	0.15	0.30	0.38
Alabama	Bullock County	10404	0.21	0.22	0.06	0.35	0.18	0.00	0.20	0.05	0.12	0.35	0.37
South Dakota	Jerauld County	2021	0.21	0.28	-0.04	0.39	0.19	0.09	0.19	0.16	0.09	0.30	0.32
Louisiana	East Carroll Parish	7279	0.21	0.20	0.08	0.35	0.17	0.01	0.22	0.12	0.15	0.24	0.29
Kansas	Seward County	23283	0.21	0.20	0.06	0.36	0.20	0.02	0.26	0.09	0.17	0.25	0.41
Louisiana	East Feliciana Parish	19534	0.21	0.19	0.06	0.37	0.18	-0.02	0.21	0.08	0.14	0.33	0.36
Georgia	Wilkinson County	9080	0.21	0.17	0.10	0.35	0.16	-0.07	0.20	0.04	0.10	0.27	0.39
Georgia	Quitman County	2277	0.21	0.26	0.03	0.33	0.18	0.11	0.21	0.03	0.11	0.32	0.27
Louisiana	West Feliciana Parish	15376	0.21	0.17	0.10	0.35	0.15	-0.03	0.19	0.06	0.11	0.21	0.34
Oklahoma	Noble County	11560	0.21	0.26	0.08	0.28	0.17	0.05	0.17	0.08	0.02	0.28	0.43
Kentucky	Ohio County	24040	0.21	0.22	0.03	0.37	0.20	0.05	0.24	0.09	0.16	0.28	0.37
Maryland	Somerset County	25705	0.21	0.16	0.12	0.34	0.15	-0.03	0.17	0.08	0.07	0.30	0.32
Ohio	Mahoning County	231988	0.21	0.20	0.06	0.36	0.15	-0.02	0.22	0.04	0.12	0.18	0.37
Texas	Refugio County	7331	0.20	0.14	0.12	0.35	0.17	0.00	0.16	0.10	0.08	0.30	0.39
Michigan	Menominee County	23494	0.20	0.19	0.11	0.31	0.14	-0.07	0.20	0.04	0.09	0.24	0.32
Nebraska	Custer County	10831	0.20	0.21	0.06	0.34	0.18	0.03	0.24	0.04	0.13	0.31	0.32
Mississippi	Jasper County	16549	0.20	0.20	0.09	0.33	0.18	0.02	0.19	0.05	0.14	0.36	0.33
Kansas	Harper County	5785	0.20	0.23	0.05	0.32	0.15	0.10	0.07	0.08	-0.02	0.23	0.43
Louisiana	St. Charles Parish	52587	0.20	0.17	0.09	0.34	0.15	-0.07	0.19	0.05	0.11	0.28	0.35
Oregon	Lake County	7789	0.20	0.18	0.20	0.24	0.07	-0.03	0.09	-0.01	-0.04	0.24	0.18
New Jersey	Salem County	63754	0.20	0.14	0.10	0.36	0.11	-0.12	0.17	0.01	0.06	0.24	0.27
Kansas	Hodgeman County	1868	0.20	0.21	0.01	0.39	0.08	0.08	0.04	0.01	0.07	0.20	0.20
Georgia	Treutlen County	6807	0.20	0.21	0.06	0.34	0.16	0.02	0.21	0.08	0.11	0.26	0.30
Mississippi	Wayne County	20515	0.20	0.20	0.06	0.34	0.16	-0.05	0.19	0.07	0.10	0.30	0.37
Louisiana	Washington Parish	46365	0.20	0.18	0.07	0.35	0.18	-0.02	0.21	0.11	0.12	0.31	0.37
Louisiana	Plaquemines Parish	23462	0.20	0.16	0.09	0.35	0.16	-0.00	0.17	0.07	0.08	0.31	0.35
Mississippi	Smith County	16073	0.20	0.20	0.14	0.26	0.17	0.03	0.16	0.12	0.08	0.33	0.30
Louisiana	Iberia Parish	73548	0.20	0.18	0.07	0.36	0.16	-0.04	0.20	0.05	0.11	0.30	0.35
Louisiana	Winn Parish	14646	0.20	0.17	0.06	0.37	0.18	-0.02	0.20	0.07	0.15	0.31	0.35
Oklahoma	Harper County	3825	0.20	0.18	0.10	0.32	0.11	-0.07	0.08	-0.10	-0.01	0.39	0.35
Minnesota	Houston County	18695	0.20	0.17	0.04	0.38	0.17	-0.01	0.25	0.03	0.16	0.17	0.41
Oklahoma	Alfalfa County	5912	0.20	0.31	-0.00	0.29	0.13	0.10	0.15	0.06	-0.06	0.15	0.39
Delaware	Kent County	173308	0.20	0.18	0.07	0.34	0.13	-0.05	0.18	0.02	0.09	0.23	0.31
Alabama	Hale County	15016	0.20	0.19	0.07	0.33	0.16	-0.01	0.17	0.09	0.09	0.28	0.32
Kentucky	Elliott County	7546	0.20	0.24	0.02	0.33	0.13	-0.03	0.23	0.03	0.12	0.23	0.23
Mississippi	Jones County	68664	0.19	0.12	0.18	0.28	0.08	-0.21	0.07	-0.04	-0.02	0.29	0.38
Louisiana	St. Bernard Parish	45436	0.19	0.17	0.09	0.32	0.14	-0.04	0.18	0.07	0.09	0.23	0.29
Kentucky	Oldham County	64586	0.19	0.13	0.08	0.38	0.15	-0.10	0.19	0.04	0.07	0.31	0.39
Oklahoma	Grant County	4497	0.19	0.25	0.06	0.27	0.10	0.05	0.09	0.12	-0.26	0.23	0.39
South Dakota	Harding County	1269	0.19	0.18	0.05	0.35	0.07	0.13	0.06	-0.02	-0.08	0.07	0.28
Texas	Newton County	14161	0.19	0.13	0.10	0.35	0.15	0.10	0.13	0.10	0.02	0.17	0.38
Louisiana	St. Mary Parish	52908	0.19	0.16	0.07	0.34	0.16	-0.04	0.19	0.08	0.10	0.24	0.35
Nebraska	Blaine County	476	0.19	-0.00	0.22	0.35	-0.01	0.01	-0.02	-0.17	-0.17	0.24	0.04
Louisiana	Morehouse Parish	26307	0.19	0.18	0.05	0.33	0.16	-0.00	0.20	0.08	0.13	0.21	0.33
Texas	Zavala County	12322	0.19	0.16	0.07	0.33	0.12	-0.08	0.19	-0.01	0.09	0.26	0.29
Texas	Trinity County	14541	0.18	0.20	0.04	0.31	0.16	-0.01	0.20	0.04	0.10	0.33	0.31
Mississippi	Quitman County	7513	0.18	0.13	0.08	0.32	0.10	-0.15	0.13	-0.01	0.02	0.36	0.26
North Dakota	Divide County	2457	0.18	0.09	0.04	0.41	0.15	0.03	0.15	0.01	0.03	0.28	0.40
Louisiana	Assumption Parish	22902	0.18	0.16	0.06	0.31	0.15	-0.01	0.18	0.07	0.09	0.22	0.32
Nebraska	Wheeler County	779	0.17	0.27	-0.06	0.31	-0.07	-0.14	-0.03	-0.05	-0.06	-0.17	0.05
Mississippi	Calhoun County	14636	0.17	0.15	0.02	0.35	0.17	-0.05	0.21	0.10	0.12	0.27	0.39
Mississippi	Amite County	12553	0.17	0.21	0.01	0.29	0.15	-0.08	0.14	0.04	0.04	0.47	0.30
Mississippi	Yazoo County	27424	0.16	0.14	0.03	0.32	0.15	-0.07	0.17	0.07	0.09	0.27	0.35
Nebraska	Arthur County	451	0.16	0.20	-0.01	0.28	-0.04	-0.14	0.08	-0.09	-0.05	-0.02	0.02
Mississippi	Jefferson Davis County	11633	0.16	0.18	0.01	0.28	0.17	0.03	0.18	0.06	0.12	0.31	0.33
Minnesota	Beltrami County	45629	0.14	0.02	0.03	0.38	0.11	-0.22	0.17	-0.02	0.04	0.17	0.50
Alaska	Kusilvak Census Area	8201	0.14	0.07	0.04	0.31	0.13	0.12	0.14	-0.01	-0.01	0.12	0.39
Michigan	Muskegon County	172506	0.14	0.08	-0.00	0.34	0.09	-0.17	0.17	-0.03	0.05	0.15	0.39
North Dakota	Slope County	783	0.14	0.28	-0.12	0.25	0.01	-0.07	0.03	-0.07	-0.05	-0.04	0.25
Mississippi	Yalobusha County	12457	0.13	0.12	0.01	0.27	0.11	-0.10	0.15	-0.03	0.04	0.30	0.31
Mississippi	Jefferson County	7507	0.13	0.08	0.01	0.30	0.09	-0.13	0.13	0.03	0.04	0.24	0.24
Arkansas	Calhoun County	5197	0.12	0.12	-0.04	0.29	0.09	-0.09	0.12	-0.03	0.05	0.27	0.24
Kansas	Lane County	1630	0.12	0.15	-0.13	0.35	0.05	0.07	0.10	-0.09	-0.06	0.08	0.18
Texas	Roberts County	906	0.12	0.19	-0.15	0.33	0.15	0.10	0.12	0.15	-0.00	0.35	0.16
Alaska	Aleutians East Borough	3373	0.12	0.10	0.01	0.24	-0.13	-0.01	-0.16	-0.24	-0.36	-0.13	0.14
Illinois	Menard County	12364	0.11	-0.06	0.07	0.32	0.01	-0.31	0.06	-0.15	-0.09	0.08	0.46
Ohio	Auglaize County	45754	0.10	0.02	-0.06	0.34	0.07	-0.21	0.15	-0.06	0.04	0.17	0.36
Alabama	Choctaw County	13237	0.10	-0.01	0.10	0.20	0.01	-0.29	-0.03	-0.11	-0.15	0.28	0.37
Virginia	Tazewell County	42930	0.09	0.07	-0.05	0.25	0.02	-0.22	0.08	-0.10	-0.02	0.18	0.22
Mississippi	Issaquena County	1337	0.05	0.12	-0.17	0.21	0.15	-0.07	0.12	0.07	0.05	0.19	0.52
Mississippi	Franklin County	7734	0.05	-0.17	0.06	0.27	-0.01	-0.36	-0.02	-0.16	-0.14	0.27	0.37

Table 2: Bottom 100 Sorted on `eneI`.

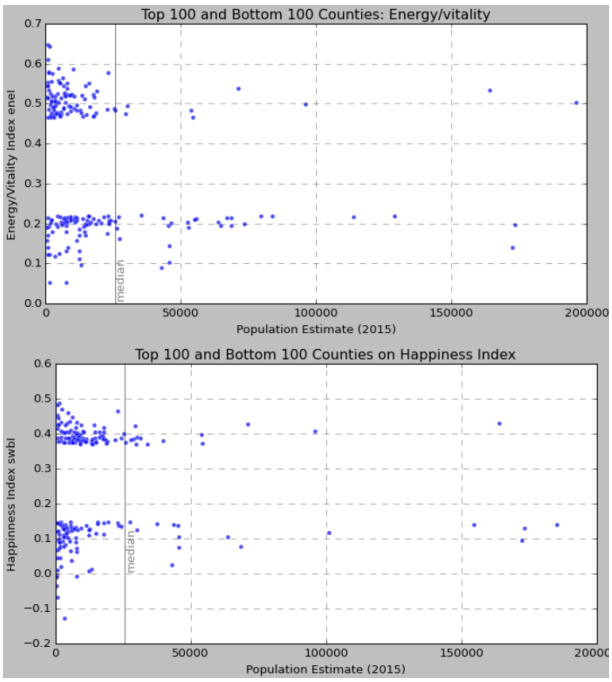


Figure 12: Highest and lowest bot energy/vitality and happiness tends to be in small counties, below median population size.